

03-09-23

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CHEMISTRY

Section (A)

1. (4) Argon
2. (2) Calcium
3. (4) 22
4. (1) 17
5. (3) 1:1
6. (4) Mercury
7. (2) Lead
8. (1) 1
9. (1) Lithium
10. (4) Ammonium Chloride
- 11.
12. (2) Consists of molecules
13. (3) An alkali
14. (1) Direct combination
- 15.

16. i)
- a) X
 - b) L
 - c) Halogens
 - d) 15 $\rightarrow$ Nitrogen
13 $\rightarrow$ Boron
 - e) O

- ii)
- a) reducing agents
 - b) bad
 - c) acidic
 - d) Silver chloride
 - e) Fluorine (F)

iii) i) No. of moles = $\frac{320}{64}$

= 5

$$5 = \frac{\text{Volume}}{22.4}$$

$$\text{Volume} = 22.4 \times 5 = 112 \text{ dm}^3$$

ii) While ~~gases~~ gases react they do so in volumes which bear a simple whole number ratio to one another and to the volume of the products, provided the temperature and pressure remain constant.

iii) No. of moles = $\frac{8.8}{44} = \frac{1}{5}$

$$\frac{1}{5} = \frac{\text{Volume of Oxygen}}{22.4}$$

$$\text{Volume of } O_2 = \frac{22.4}{5} = 4.48 \text{ dm}^3$$

iv) i) $\ln 23g = 6.022 \times 10^{23} \text{ atoms}$

$\ln 1g = \frac{6.022 \times 10^{23}}{23}$

$\ln 4.6g = \frac{6.022 \times 10^{23}}{23} \times 4.6$

$= 6.022 \times 10^{22} \times 5$

$= 30.611 \times 10^{22} \text{ atoms}$

ii) $RMM = 64 + 32 + 64 + (2 + 16)5$

$= 160 + 90$

$= 250$

Percent of $SH_2O = \frac{90}{250} \times 100$

$= 18 \times 2$

$= \underline{36\%}$

v) i) - 15

ii) - 19

iii) - 8

iv) - 4

v) - 2

Section (B)

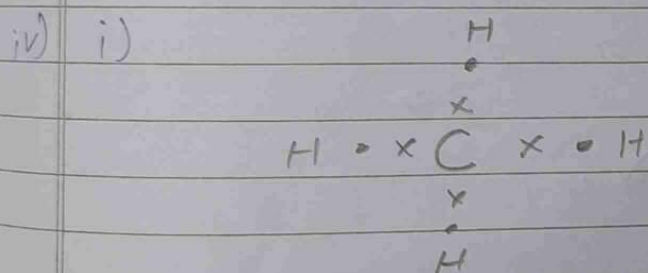
1. i) True. Cations receive electrons and

2. i) False. Anion receive electrons and gain the configuration of the noble gas after them.

2. True. Electrovalent compounds are good conductors in their aqueous / molten state.

$$\begin{aligned}
 \text{ii) No. of moles} &= \frac{22}{\text{GMM}} \\
 &= \frac{22}{12+16+16} \\
 &= \frac{22}{44} = \frac{1}{2} \text{ moles} \\
 &= 0.5 \text{ moles}
 \end{aligned}$$

- iii) a) It is greater than Y
 b) It is greater than Y
 c) To the right of Y



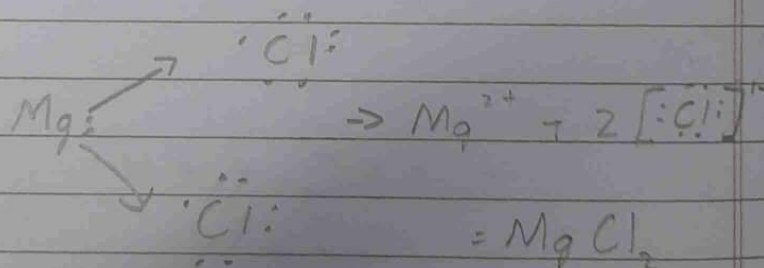
- ii) a) Ba as its ionization potential is highest
 b) They have two electrons in their outermost shell.

3. i) Sulphur - It forms MS
~~Chlorine~~ Chlorine - It forms MCl_2

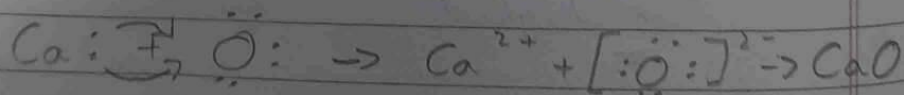
ii) a) This is because Hydrogen and Oxygen each share a single electron and hence, a single covalent bond. While ~~two~~ Oxygen atoms each share a pair of electrons and hence, a double covalent bond.

b) A molecule of CCl_4 has 4 single bonds as a carbon atom shares one atom each with 4 chlorine atoms hence, there are 4 single ~~one~~ bonds.

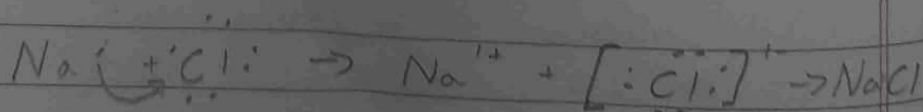
iii) i) MgCl_2



ii) CaO



iii) NaCl



iv) A bond in which one of the combining atoms contributes both of the shared electrons.

Eg. NH_4Cl

4 i) a) 2, 8, 8, 2

b) Group 2, period 4

ii) a) 2^+

b) Ionic compound

iii) Avogadro's law states that -
Under the same conditions of temperature and pressure, equal volumes of all gases contain the same number of molecules.

Applications are:-

- It helps determine atomicity
- It helps determine molecular formula

iv) i) A & B

ii) B & D

iii) B & D

6.i) a) Copper sulphate $\rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

b) Potassium nitrate $\rightarrow \text{KNO}_3$

ii) Deliquescence $\rightarrow$ The process of absorbing ^{moisture} water from the atmosphere and changing state

Efflorescence $\rightarrow$ The process of losing water when exposed to the atmosphere

- iii) • Acids neutralize bases to form a salt and water.
• React with active metals to form hydrogen

- iv) a) Acid
b) Base
c) Base
d) Acid
e) Acid
f) Acid